Fiscal Year:	FY 2011	**Task Last Updated:**	FY 05/11/2011
PI Name:	Strangman, Gary E Ph.D.		
Project Title:	Distributed system for spaceflight biomedical support		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Smart Medical Systems and Technology Team		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **ExMC**:Exploration Medical Capabilities		
Human Research Program Risks:	(1) **Medical Conditions**:Risk of Adverse Health Outcomes and Decrements in Performance Due to Medical Conditions that occur in Mission, as well as Long Term Health Outcomes Due to Mission Exposures		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	strang@nmr.mgh.harvard.edu	**Fax:**	FY
PI Organization Type:	NON-PROFIT	**Phone:**	617-724-0662
Organization Name:	Massachusetts General Hospital		
PI Address 1:	Department of Psychiatry		
PI Address 2:	149 13th Street, Suite 2651		
PI Web Page:			
City:	Charlestown	**State:**	MA
Zip Code:	02129-2020	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Crew Health NNJ09ZSA002N
Start Date:	06/01/2010	**End Date:**	05/31/2014
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Zeffiro, Thomas (Argosy Omnimedia, Inc.) Zhang, Quan (Massachusetts General Hospital) Montgomery, Rob (Argosy Omnimedia, Inc.)		
Grant/Contract No.:	NCC 9-58-SMST02201		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Over the past three decades, scores of biomedical monitoring devices have been deployed aboard U.S. spacecraft and the International Space Station. These have enabled collection of sleep/wake activity, temperature, heart rate, respiration rate, and ultrasound images, among other parameters. However, nearly all such measurement systems have been stand-alone devices, unable to operate together, synchronize data streams, be easily augmented with new sensors, or coordinate in any way with therapeutic devices. Although some systems under development can record multiple signals, even these merged systems remain stand-alone devices. Currently, there is no flight-ready platform that can (a) bring together data from disparate human and environment sensors for diagnosis, (b) make acquired data available from nearly any location or platform, or (c) intelligently incorporate therapeutic components of the spaceflight medical system. Our ultimate objective is to develop a prototype platform, called SpaceMed, that can seamlessly integrate disparate biomedical devices, including future therapeutic and decision-support systems. SpaceMed software will consist of three primary components: (1) listener services that probe the external environment for both wired devices (e.g., USB, audio/video feeds) and wireless devices (e.g., bluetooth, radio, WiFi), (2) networking publish/subscribe middleware that enables any data or control signals that are "published" to be automatically delivered to any interested receiving devices (as well as being archived), and (3) graphical interface software for display of the collected data and to facilitate device control. In addition to developing the SpaceMed software, we will develop demonstration hardware devices to integrate with SpaceMed. Thus far, we have integrated two commercial-off-the-shelf (COTS) hardware platforms with SpaceMed, using a variety of communication capabilities: Bluetooth, 802.15.4 radio communication (both ANT and TinyOS standards), and USB mass storage devices. The available devices--including sensors for ECG, heart rate, temperature, accelerometry, environmental carbon dioxide, and USB memory sticks--are automatically discovered, connections are created, and data is acquired and loaded into the middleware without human intervention. We plan to demonstrate support for the current ISS carbon dioxide monitors (CDMs) and the next generation ultrasound system, and are working on integrating a near-infrared neuroimaging (NIN) system capable of recording brain or tissue oxygenation. This development process will also help us develop adapter hardware and an associated process for retrofitting existing medical devices. Given this platform design, future additions to the medical kit should require no more than a supported communication capability for sending and/or receiving device data to integrate with the SmartMed system. Our project will thus (1) provide a prototype but standardized platform for future medical capabilities integration, (2) leverage previous NASA/NSBRI funding from NASA Ames, NASA Glenn, and the NIN devices, (3) demonstrate the feasibility of seamless interoperation of such components without human intervention, (4) validate the approach for integrating telemetrically gathered physiological data streams from multiple devices and display time-synchronized data from multiple nodes as the information is acquired, and (5) demonstrate an upgrade path to connect future, not-yet-developed medical kit components with the system.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Impact: The integrated platform and devices, plus the autonomous operation of SpaceMed, will provide consolidated data from throughout the spacecraft--previously unavailable to crews in-flight--while avoiding data consolidation burdens currently placed on crew members. Real-time integration of medical and environmental data and devices will facilitate early and efficient detection and treatment of medical conditions. For example, real-time access to ambient CO2 levels in ISS nodes alongside multi-astronaut cardiac and respiratory data could motivate proactive CO2 scrubbing activities. Or, periodic measures of optic nerve sheath diameter and cognitive performance could enable early identification of intracranial hypertension. SpaceMed could also potentially reduce training time by automating or consolidating medical system complexity. Integration of diagnostic and therapeutic elements will be essential for developing more autonomous medical systems for exploration-class missions. Earth Benefits: We expect SpaceMed will have substantial spin-off impact in Earth-based settings, as there are similar medical data acquisition and consolidation needs in numerous hospital and first-responder settings. Benefits are expected to range from efficiency improvements in intensive care units, where monitors are constantly swapped in and out, to deployment in military and first responder settings where a lightweight, robust, distributed and easily upgradable architecture is essential.
Task Progress:	During the first year of our project developing the SmartMed platform, we have made progress in hardware, software and graphical interface development. To demonstrate the ability to integrate with different commercial wireless devices and protocols, we have deployed both standard and developed custom wireless hardware for use with the SpaceMed platform. Standard sensors that we have incorporated include electrocardiography (ECG), electromyography (EMG), galvanic skin response (GSR), accelerometry, gyroscopes, and temperature sensors. Custom sensors have included both Bluetooth and 802.15.4 radio environmental carbon dioxide sensors. Support for data acquisition from any USB-based mass storage device was also achieved. Jointly, this collection of hardware demonstrates both medical and environmental sensing, using devices based on both wireless and wired communication and transfer protocols. We then developed our initial listener services. The first four modular listener services can acquire data from the hardware describe above, including: (i) USB mass storage devices, (ii) Bluetooth wireless sensors devices, and 802.15.4 radio devices through either the (iii) TinyOS or the (iv) ANT protocols. The listener services feed their data streams to BioNet middleware, which has been adopted for version 1.0 of the SmartMed platform. This middleware allows data acquired from the listener services to be "published" to the BioNet virtual domain. Any client connected to this domain (e.g., databases, sensor monitoring stations, mission control, or flight surgeon mobile devices) will be allowed to "subscribe" to any or all data available on BioNet, restricted by authenticated logins and access authorization tables. Following extended interaction with the ExMC element--including a thorough evaluation of the ExMC Gap Analysis requirements, needs and desires--we have begun designing version 1.0 of the SmartMed graphical interface. The GUI provides a dockable interface for display flexibility, and can be accessed via thin client (web browser), enabling full portability across all computing platforms (Windows, Linux, Mac, smart phones). Once complete, the SmartMed platform will provide a highly automated way to acquire, access and store data during spaceflight. Improvements in access will be particularly notable in-flight, where essentially no such access currently exists. The SmartMed platform's capabilities will aid in early recognition and identification of clinical pathologies (ExMC gaps 1.01, 3.03), provide a simpler path to integrate new monitoring capabilities (ExMC gap 4.19), automate

	much of the medical data collection and management process (ExMC gaps 4.20, 5.01), as well as provide an integrated platform to interface with decision assistance and closed-loop systems (ExMC gap 4.01). Importantly, these capabilities will be provided while simultaneously reducing crew time required for data management and system administration.
Bibliography Type:	Description: (Last Updated: 07/08/2026)
Abstracts for Journals and Proceedings	Strangman GE, Montgomery RR, Jamieson IMD. "Distributed system for spaceflight biomedical support." 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. 18th IAA Humans in Space Symposium, Abstract book, April 2011. , Apr-2011